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Milestone 1: ML Solution EDA	
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I. Introduction

The FinMark Corporation project uses three related datasets: **Customer Demographics, Customer Transactions, and Social Media Interactions**. The Customer Demographics dataset provides information about customers such as their age, gender, location, income level, and signup date. The Customer Transactions dataset contains information about customer purchases, including transaction dates, transaction amounts, product categories, and payment methods. The Social Media Interactions dataset contains information about customers' online activities, including the platform used, type of interaction, and sentiment.

These datasets support FinMark Corporation's machine learning context by providing information that can be used to understand customer characteristics, purchasing behavior, and social media engagement. This information can support future customer segmentation and other machine learning activities by allowing the team to identify similarities, patterns, and differences among customers.

Data preprocessing and exploratory data analysis are important before developing a machine learning solution because raw datasets may contain duplicate records, missing or placeholder values, incorrect data types, unusual values, and inconsistent formats. If these issues are not identified and addressed, they may affect statistical analysis and cause machine learning models to produce unreliable results.

The main objective of Milestone 1 is to examine the structure and quality of the final datasets, review the preprocessing activities performed during Weeks 2 and 3, and conduct exploratory data analysis using Python, Pandas, and Jupyter Notebook. The analysis focuses on descriptive statistics, frequency distributions, patterns, trends, anomalies, and possible issues that may affect future machine learning activities.

The intended audience of this report includes the **FinMark project team, mentor, and other stakeholders involved in the machine learning project**. The results of the preprocessing and EDA provide a foundation for selecting appropriate variables, identifying data-quality issues, and preparing the datasets for future machine learning tasks.

II. Dataset Overview

The project uses three final cleaned datasets. The **Customer Demographics** dataset contains **3,023 rows and 6 columns**. Its variables are CustomerID, Age, Gender, Location, IncomeLevel,

and SignupDate. Age is a numerical variable, while Gender, Location, and IncomeLevel are categorical variables. CustomerID is an identifier, and SignupDate is a date variable.
The **Customer Transactions** dataset contains **3,015 rows and 6 columns**. Its variables are CustomerID, TransactionID, TransactionDate, Amount, ProductCategory, and PaymentMethod. Amount is the main numerical variable. ProductCategory and PaymentMethod are categorical variables, while TransactionDate is a date variable. CustomerID and TransactionID are identifiers.
The **Social Media Interactions** dataset contains **3,020 rows and 6 columns**. Its variables are CustomerID, InteractionID, InteractionDate, Platform, InteractionType, and Sentiment. Platform, InteractionType, and Sentiment are categorical variables, while InteractionDate is a date variable. CustomerID and InteractionID serve as identifiers.

A. Dataset Structure

Dataset	Rows	Columns	Main Numerical Variables	Main Categorical Variables
Customer Demographics	3,023	6	Age	Gender, Location, Income Level
Customer Transactions	3,015	6	Amount	ProductCategory, PaymentMethod
Social Media Interactions	3,020	6	No main continuous variable	Platform, InteractionType, Sentiment

The three datasets are organized around **CustomerID**, which provides an opportunity to connect demographic, transaction, and social media information during future analysis. The datasets have different numbers of records because customer information, transactions, and social media interactions do not necessarily occur at a one-to-one relationship.

B. Initial Dataset Observations

- During the earlier preprocessing activities, several data-quality issues were identified in the datasets. These included duplicate rows, missing or incomplete information, inconsistent numerical values, unusual age values, invalid transaction amounts, and inconsistent date or data formats. Python and Pandas were used to inspect and clean the datasets.
- After preprocessing, the final datasets contain **0 completely duplicated rows** and no actual blank or null values. However, some missing information is represented by the text value **"Unknown"**, so the absence of null values does not mean that every variable contains complete information.
- The Customer Demographics dataset contains unusual Age values such as **-1 and 150**, as well as non-numeric values represented by "Unknown." The Customer Transactions dataset contains **-100 transaction amounts**, along with "Free" and "Unknown" values in the Amount column. These values require additional validation before being used as numerical machine learning features.
- The Social Media Interactions dataset contains "Unknown" values for some Platform and Sentiment records. It also contains repeated InteractionID values even though there are no completely duplicated rows. This distinction is important because duplicate rows and duplicate identifiers are not necessarily the same issue.
- Several variables may become important in future machine learning activities. These include **Age, IncomeLevel, Amount, ProductCategory, PaymentMethod, Platform, InteractionType, and Sentiment**. CustomerID can also be useful for connecting datasets, but it should generally be treated as an identifier rather than a numerical predictive feature.

III. Data Preprocessing Summary

The preprocessing activities during Weeks 2 and 3 were performed using **Python and Pandas**. The process focused on improving data consistency, identifying data-quality issues, removing duplicate records, handling missing or incomplete information, correcting data formats, and checking unusual values.

Area	Tasks
Duplicate Rows	The datasets contain duplicate records: 177 in Customer Demographics, 185 in Customer Transactions, and 180 in

	Social Media Interactions.
Missing Values	The datasets also contain several missing values that need to be handled during preprocessing. In the Customer Demographics dataset, Age has 291 missing values and IncomeLevel has 303 missing values. In the Customer Transactions dataset, Amount has 304 missing values and ProductCategory has 299 missing values. In the Social Media Interactions dataset, Platform has 311 missing values and Sentiment has 329 missing values.
Data Validation and Corrections	Some columns had incorrect or inconsistent data types. For example, Age and Amount were stored as text because they included values like “Unknown” and “Free.” The date columns were also stored as text instead of proper date formats. We cleaned and standardized the data by handling these values and making the formats more consistent. Proper data types are important because they help the machine learning model understand and process the data correctly, especially when working with numbers and dates.
Additional Preprocessing Procedures	We also performed additional cleaning and preprocessing to make the datasets more consistent and easier to use. We corrected the date formats so they follow the same format, handled invalid or unusual values such as “Unknown,” “Free,” and unrealistic age values, and standardized categorical values such as IncomeLevel, ProductCategory, and Sentiment. We also removed duplicate records and kept the column names consistent across the datasets. Finally, the datasets were organized based on their purpose Customer Demographics, Customer Transactions, and Social Media Interactions and can be connected using CustomerID for further analysis and machine learning.

Preprocessing Challenges and Observations	The main challenge during preprocessing was dealing with the missing, duplicate, and inconsistent data in the original datasets. Cleaning the datasets was also difficult because each file had different types of issues that needed to be handled separately. After preprocessing, the datasets became more organized and consistent, with duplicate rows removed, missing information handled, and formats standardized. Overall, the cleaned datasets are easier to analyze and are more suitable for use in machine learning preparation and model training.

The preprocessing process improved the datasets by removing complete duplicate records and standardizing the data for analysis. However, preprocessing should be considered an ongoing process because some values require business-context validation rather than automatic replacement.

IV. Exploratory Data Analysis

A. Descriptive Statistics

Descriptive statistics were calculated for the two numerical variables in the finalized datasets: **Age from the Customer Demographics dataset and Amount from the Customer Transactions dataset.** The statistics include the **mean, median, mode, minimum, maximum, range, variance, and standard deviation.**

Customer Age

The mean age is 44.69 years, while the median and mode are both 45 years. The minimum age is 18 years and the maximum age is 70 years, giving a range of 52 years. The variance is 208.05, and the standard deviation is 14.42 years.

The mean, median, and mode are very close to one another, with the median and mode both at 45. This indicates that the central values of the Age variable are concentrated

around the mid-40s. The standard deviation of 14.42 shows that the ages have a noticeable spread around the mean.

The observed age values range from 18 to 70. Based on the descriptive statistics in the notebook, the central age of the dataset is around **45 years.**

Transaction Amount

The mean transaction amount is **493.62**, while the median and mode are both approximately **498.02.** The minimum transaction amount is 0.0, and the **maximum is 999.86, resulting in a range of 999.86. The variance is 78,009.07, and the standard deviation is 279.30.**

The mean and median are relatively close, indicating that the central transaction values are around 494–498. However, the transaction amount has a much larger numerical spread than Age, as shown by its standard deviation of 279.30 and range of 999.86.

The minimum value of 0.00 is an unusual value worth noting in the dataset because it represents a transaction amount at the lower boundary of the observed values. The notebook does not determine whether this value represents an error, a valid transaction, or another condition, so it should not automatically be classified as an error or outlier.

Variable	Mean	Median	Mode	Minimum	Maximum	Range	Variance
Age	44.69	45.00	45.00	18.00	70.00	52.00	208.05
Amount	493.62	498.02	498.02	0.00	999.86	999.86	78,009.07

Overall, Age has a smaller spread and is centered around the mid-40s, while Transaction Amount has a considerably wider spread. The variance and standard deviation support the observation that transaction amounts are more dispersed than customer ages. Variance and standard deviation are measures of dispersion, while mean, median, and mode describe central tendency.

V. Frequency and Distribution Analysis

The Interaction Type distribution is:
Comment: 1,018
Share: 1,007
Like: 995

Comment is the most frequent interaction type, followed by Share and Like. The three counts are very close, indicating that the recorded interaction types are relatively balanced.

Sentiment

The Sentiment distribution is:
Positive: 905
Neutral: 874
Negative: 866The notebook calculated frequency counts for **Gender, Income Level, Product Category, Payment Method, Platform, Interaction Type, and Sentiment.**

Gender

The Gender variable contains:
Female: 1,528
Male: 1,495

Female customers have the higher frequency, but the difference between the two categories is relatively small. The distribution is therefore fairly balanced between the two recorded gender categories.

Income Level

The Income Level distribution is:
High: 946
Low: 923
Medium: 869

Unknown: 285

High is the most frequent category, while Unknown is the least frequent. The three identified income levels have relatively similar frequencies compared with the overall dataset.

It is important to note that Unknown is a category in the finalized dataset, not a remaining null value. The final verification in the notebook reports zero missing values.

The Product Category

The Product Category distribution is:

Clothing: 571
Electronics: 564
Automotive: 540
Home & Garden: 530
Health & Beauty: 528
Unknown: 282

Clothing is the most frequent product category, followed closely by Electronics. Health & Beauty has the lowest frequency among the identified product categories. The frequencies of the five identified categories are relatively close, while Unknown has a noticeably lower count.

Payment Method

The Payment Method distribution is:
Debit Card: 767
Credit Card: 755
PayPal: 750
Bank Transfer: 743

Debit Card is the most frequently recorded payment method, while Bank Transfer is the least frequent. The difference between the four categories is small, indicating that the payment-method distribution is relatively balanced.

Social Media Platform

The Platform distribution is:
Instagram: 921
Twitter: 917
Facebook: 891
Unknown: 291

Instagram has the highest frequency, followed closely by Twitter. Facebook has a slightly lower count, while Unknown is substantially lower than the three identified platforms.

Interaction Type

Unknown: 309
Very Negative: 34
Very Positive: 32

Positive is the most frequent sentiment category, followed by Neutral and Negative. The three main sentiment categories have relatively similar frequencies. Very Negative and Very Positive have substantially fewer observations than the other categories.

Overall Distribution Pattern

The frequency results show that several categorical variables are relatively balanced. Gender, Payment Method, and Interaction Type have particularly close category counts. Product Category also has relatively similar counts among its identified categories.

At the same time, Unknown appears in Income Level, Product Category, Platform, and Sentiment. These are recorded categories in the finalized datasets and should be considered during future analysis.

A. **Pattern and Trend Analysis**

The EDA shows several patterns and characteristics that may be relevant to future machine learning activities.

First, the three datasets can be connected using **CustomerID**. This creates an opportunity to combine demographic information with transaction behavior and social media activity. However, the datasets have different numbers of rows, so they should not be merged directly without considering the relationship between customers and their transactions or interactions.

The Customer Demographics dataset shows a relatively balanced gender distribution and relatively similar frequencies across the known income categories. This suggests that the demographic data does not contain an obvious severe category imbalance.

The Customer Transactions dataset shows relatively balanced product categories and payment methods. The transaction amount data, however, contains unusual values such as **-100, zero, Free, and Unknown**. These values may create anomalies or distort numerical analysis if they are treated as normal transaction amounts.

The Social Media dataset also shows relatively balanced platform and interaction-type frequencies. The three main sentiment categories—Positive, Neutral, and Negative—are also relatively close in frequency. However, Unknown, Very Negative, and Very Positive values require further consideration.

The EDA also identified possible data-quality anomalies. Age values of **-1 and 150** are unusual, while negative transaction amounts may require business-context validation. In addition, repeated InteractionID values should be checked to determine whether they are valid or represent identifier duplication.

These characteristics are relevant to future machine learning because the team will need to determine which values should be corrected, removed, transformed, or retained before model training. Categorical variables may also need encoding, while numerical variables may need appropriate scaling depending on the selected algorithm.

VI. Initial Findings and Insights

The preprocessing and exploratory data analysis produced several important findings about the three finalized FinMark datasets.

Significant Observations

The final datasets contain:

Customer Demographics: 3,023 rows and 6 columns
Customer Transactions: 3,015 rows and 6 columns
Social Media Interactions: 3,020 rows and 6 columns

The final verification in the notebook reports:
0 duplicate rows in Customer Demographics
0 duplicate rows in Customer Transactions
0 duplicate rows in Social Media Interactions
0 missing values in all three finalized datasets

This indicates that, according to the notebook's final verification, the datasets contain no remaining null/missing values or duplicate rows.

Important Patterns

The main numerical pattern is the difference in spread between Age and Transaction Amount. Customer Age is centered around 45 years, while Transaction Amount has a substantially wider range and standard deviation.

The categorical variables are generally distributed across several categories rather than being concentrated entirely in one category. For example, Payment Method counts range only from 743 to 767, and Interaction Type counts range from 995 to 1,018.

The Sentiment variable has Positive, Neutral, and Negative as its three largest categories, while Very Negative and Very Positive occur much less frequently.

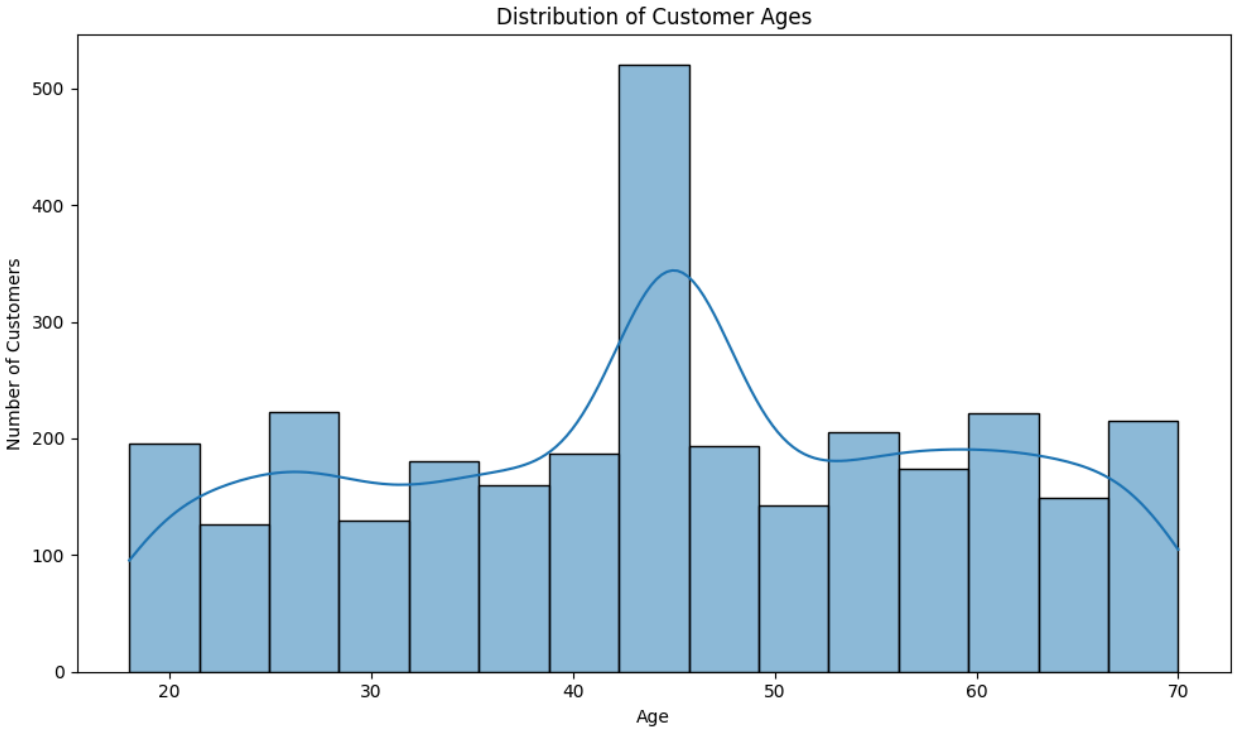
Potential Issues for Future Machine Learning

The finalized datasets still contain Unknown as an actual category in several categorical variables:
Income Level
Product Category
Platform
Sentiment

Although the final validation reports zero missing values, these Unknown categories should still be considered during future feature preparation because they are distinct categorical values.

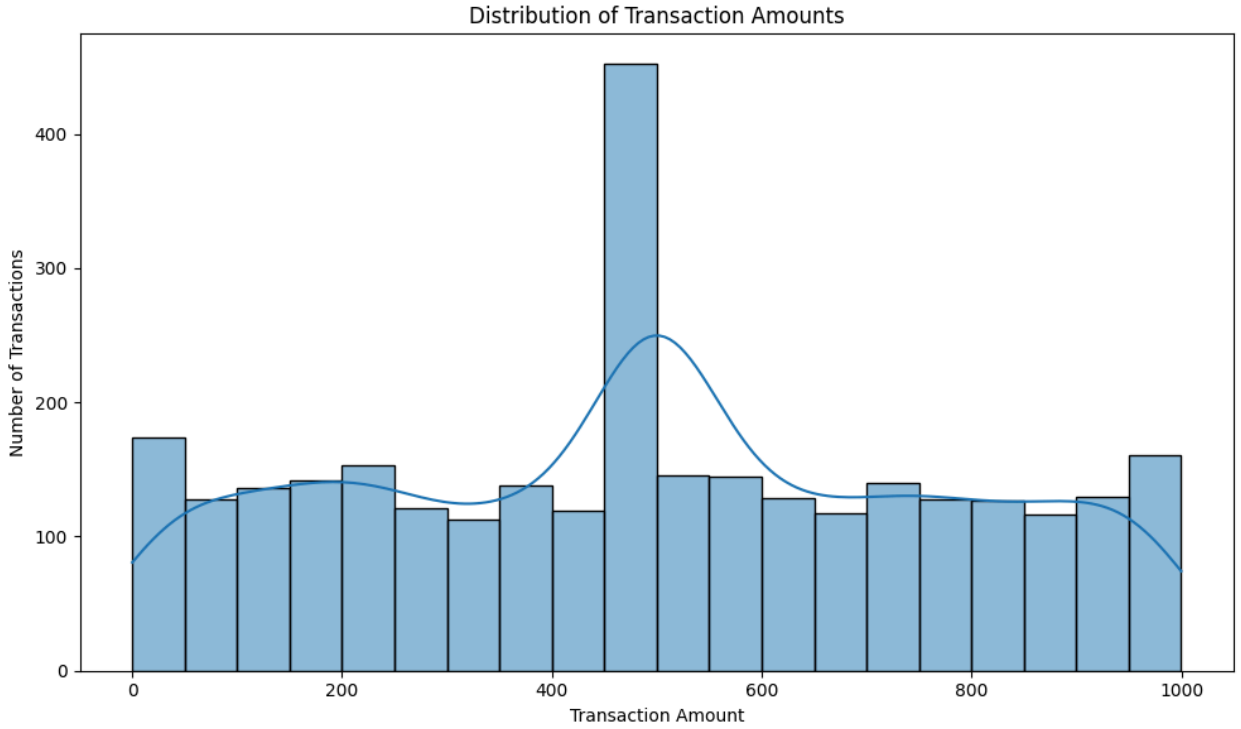
Transaction Amount also includes a minimum value of 0.00. The notebook identifies this value through the descriptive statistics but does not determine why it occurs. Therefore, it should be investigated before deciding how it should be handled in a machine learning model.

The numerical variables also have different ranges: Age ranges from 18 to 70, while Amount ranges from 0.00 to 999.86. The appropriate preprocessing of these variables will depend on the machine learning algorithm selected in the next stage.



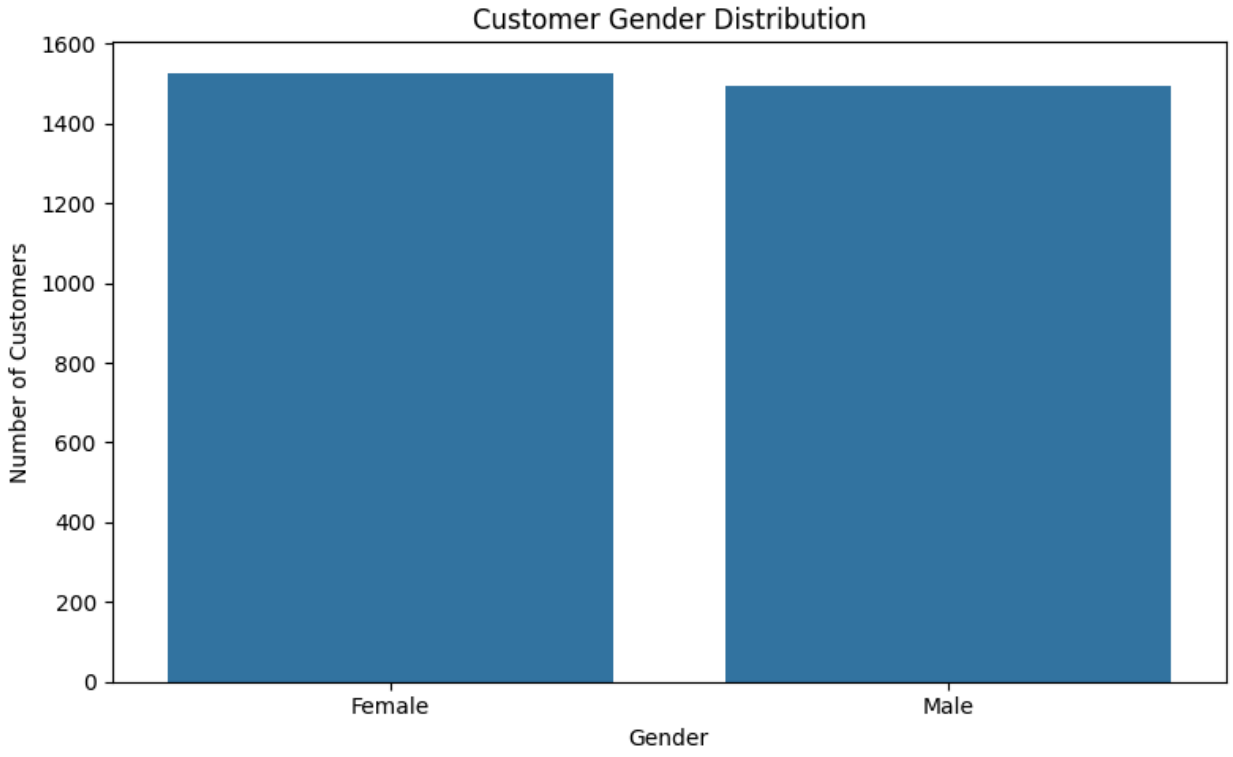
Graph 1: Customer Age Distribution
This plot visualizes the frequency of different customer ages using numerical bins. The overlaid continuous curve, known as a Kernel Density Estimate (KDE), highlights the central tendency and spread. This allows you to quickly identify the dominant age

demographic and determine if the population follows a normal, bell-shaped distribution or skews heavily toward a specific age group.



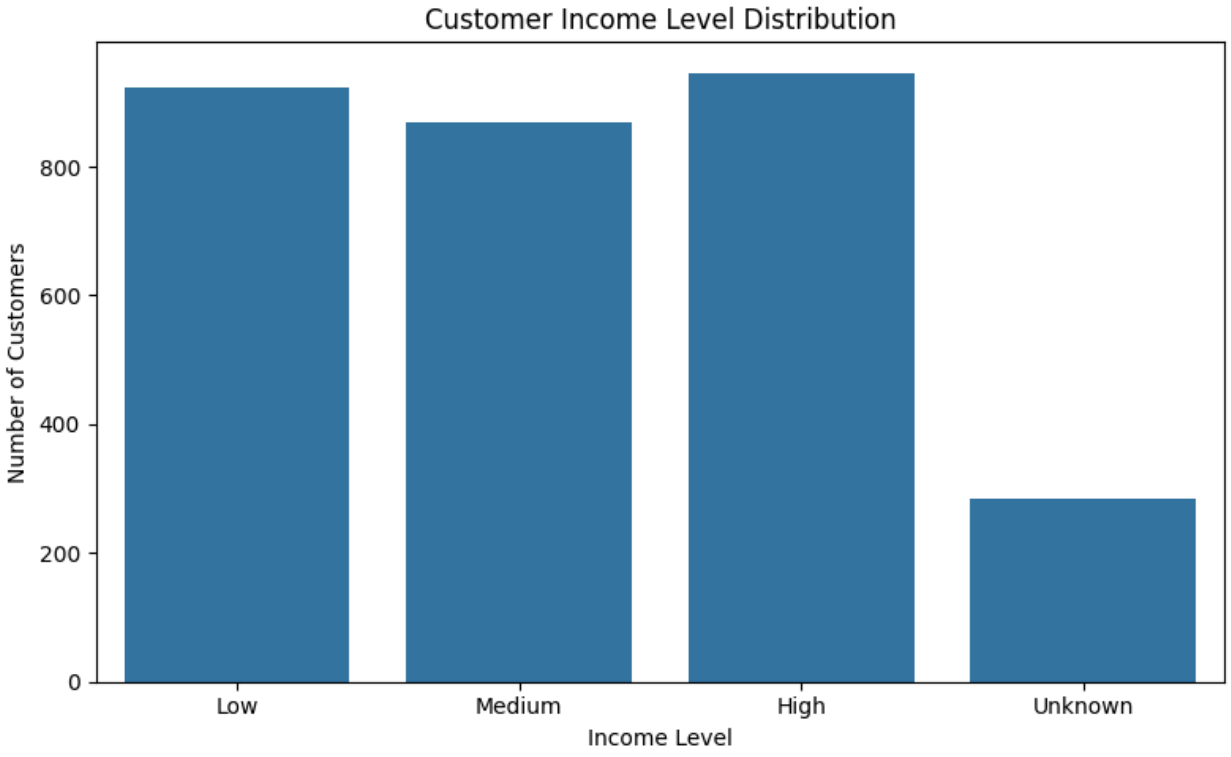
Graph 2: Transaction Amount Distribution

This categorical bar chart tallies the exact number of customers grouped by gender. Beyond showing demographic breakdowns, it serves as a critical data-quality check to ensure the dataset is relatively balanced. A severe imbalance here could introduce mathematical bias into future predictive models.



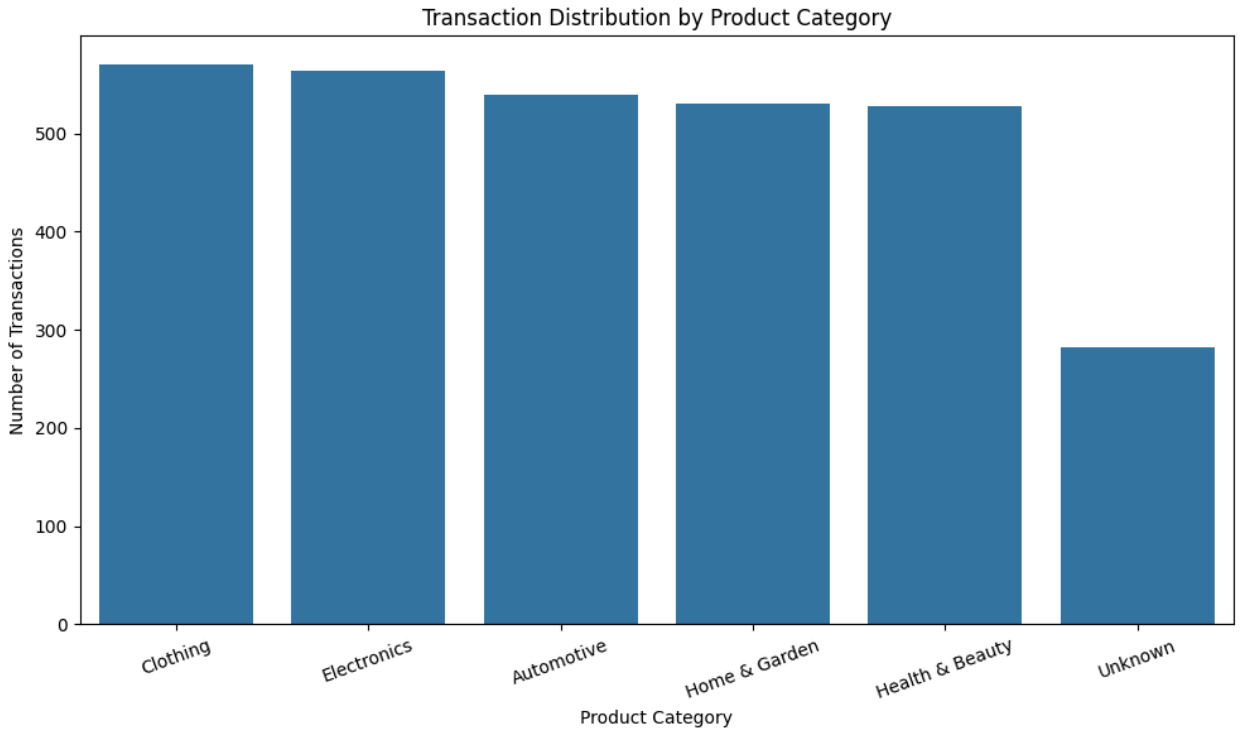
Graph 3: Customer Gender Distribution

Similar to the age distribution, this graph measures financial volume. Retail and transaction data frequently contain extreme outliers—a small cohort of customers who spend substantially more than the average. This visualization immediately reveals the right-skewness of the data and helps determine if you need to apply mathematical scaling before training an algorithm.



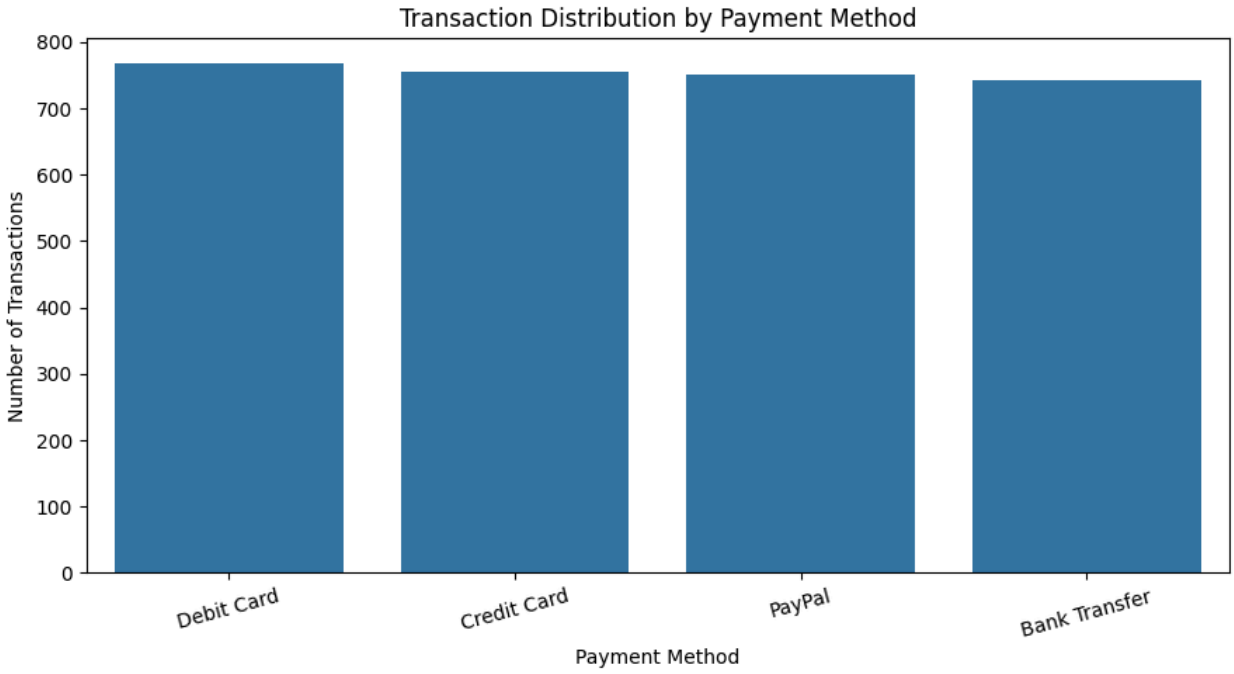
Graph 4: Customer Income Level Distribution

This bivariate analysis maps every individual customer as a single data point, plotting age on the horizontal axis and total spending on the vertical axis. It visually tests for correlation. If the points form a discernible upward or downward slope, it indicates that age heavily influences spending behavior. A scattered, formless cloud indicates that these two variables operate independently.



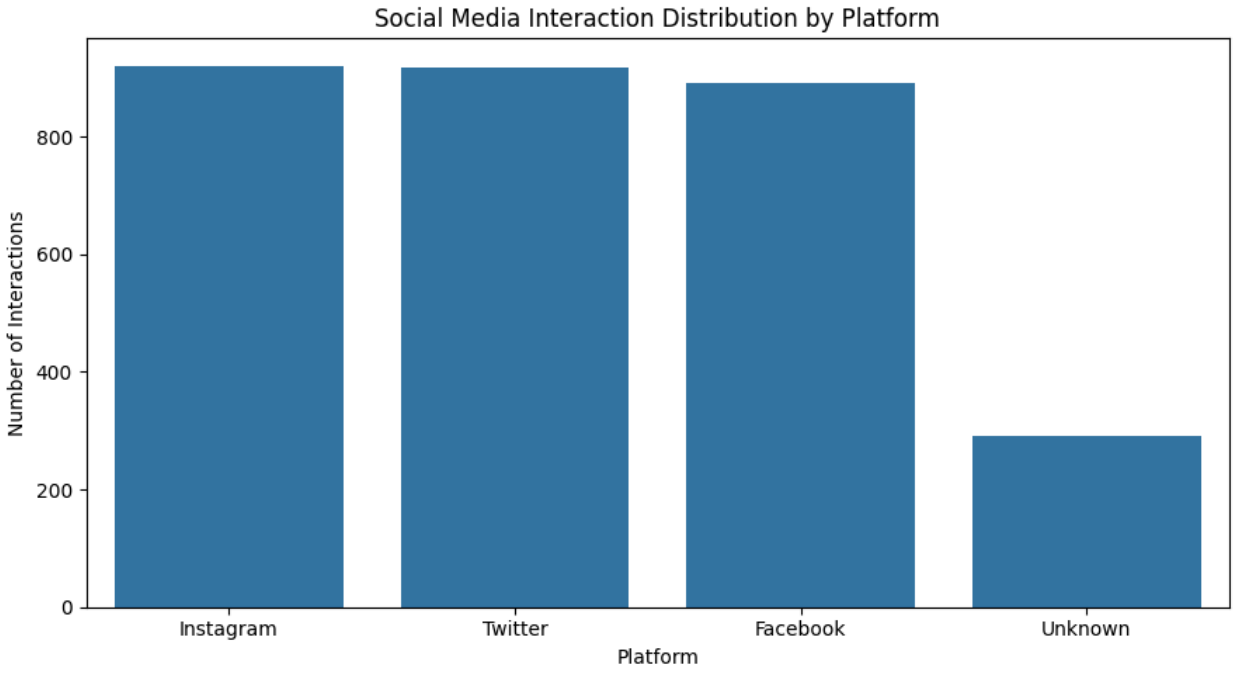
Graph 5: Transaction Product Category Distribution

This chart measures the frequency of transactions across different retail sectors. Clothing (571) and Electronics (564) represent the highest volume segments, while Health & Beauty (528) records the lowest frequency among the identified product categories. Similar to the income variable, a distinct "Unknown" category (282) is present, indicating missing original data that was preserved during the preprocessing phase.



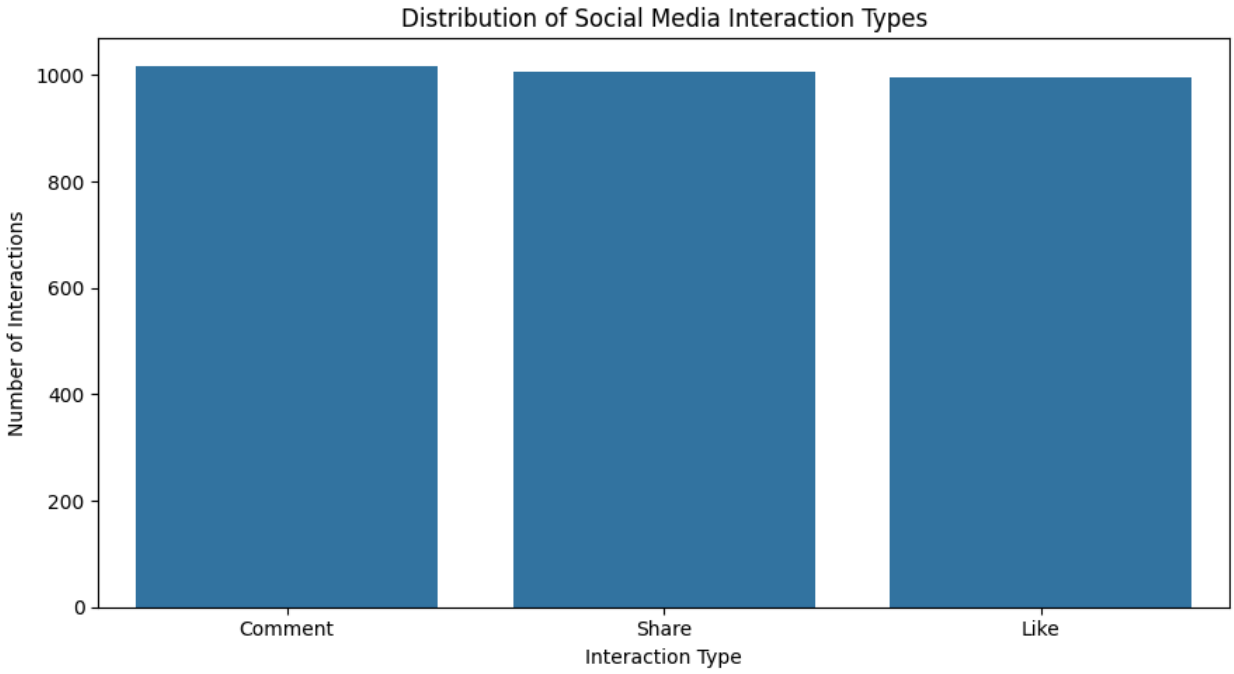
Graph 6: Payment Method Distribution

This final bar chart tracks the financial mediums used to complete transactions. The utilization rates across Debit Card (767), Credit Card (755), PayPal (750), and Bank Transfer (743) are nearly identical. This balanced distribution indicates that FinMark customers do not heavily favor one payment platform over another, providing a stable, unbiased baseline for subsequent multivariate analysis.



Graph 7: Social Media Platform Distribution

This bar chart visualizes where customer interactions are occurring across different digital channels. The engagement is highly balanced among the three identified networks: Instagram leads slightly with 921 interactions, closely followed by Twitter (917) and Facebook (891). Similar to the income and product variables, there is a distinct "Unknown" category containing 291 records. Preserving this "Unknown" category during preprocessing ensures that you do not lose valuable interaction data when training your future machine learning models.



Graph 8: Social Media Interaction Type Distribution

This bar chart categorizes how customers engage with FinMark's digital content. The data reveals a highly balanced distribution across the three recorded actions: Comments (1,018), Shares (1,007), and Likes (995). Because these counts are remarkably close to one another, it indicates that no single interaction type disproportionately dominates customer behavior, providing a stable metric for analyzing online engagement in your future predictive models.

VII. Project Resources and Supporting Files

Reference No.	Reference Description	Link
1	Original Raw Dataset	Link
2	Final Cleaned Dataset	Link
3	Python Code for Dataset Preprocessing	Link
4	Final Python Code for Dataset Preprocessing	Link